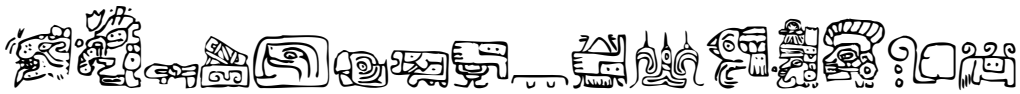
INDIAN INSTITUTE OF TECHNOLOGY BHUBANESWAR

B.Tech. Physics: Autumn 2023

Dia: Today! March 2023 **PH1L001** Time: uma hora



Practice Problems - I : Potential Formulation and Phase Diagrams MM:20

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1. A particle is under the influence of a force $F = -kx + kx^3/a^2$, where, k and a are constants and k is positive. Determine $U(x)$ and discuss the motion at various points of the potential[1]. What happens when $E = \frac{1}{4}ka^2$? (5)

2. (a) Give a plot of the following potential and discuss the classical motion of a particle moving in the potential (3)

$$U(x) = ax - bx^3 \ .$$

where a, b are positive constants. You can assume suitable values for plotting.

(b) If a particle of mass m is at a stable equilibrium position of the above potential energy, what is the minimum velocity that has to be imparted to the particle, to render its motion unstable? (2)

3. The total energy of a one-dimensional system is given as $E = \frac{1}{2}\,m\,\dot{x}^2 + V(x)$, where, m is constant, the mass of the particle, and $V(x)$ is the potential. Given below are phase diagrams (velocity/momentum vs displacement plot) for two systems: Figure-1: $V(x) = \frac{1}{2}\,k\,x^2$, where, k is a constant corresponding to a simple harmonic oscillator (Different ellipses are for different values of E) and Figure-2: $V(x) = 0$, corresponding to a free particle. The arrows indicate the direction of time flow.

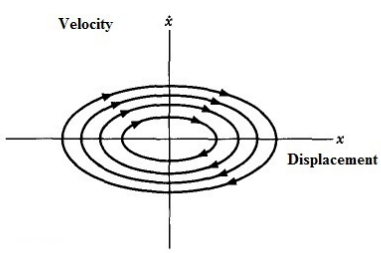

Figure-1: Phase Diagram for a Simple Harmonic Oscillator

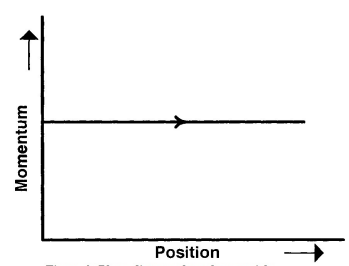

Figure-2: Phase diagram for a free particle

(a) If there are frictional forces (damping) in the system, how will the phase diagrams in Figure-1 and Figure-2 change? Draw the modified diagrams.

(b) Plot $V(x) = k\,x$ where, k is a constant. Give an example of a physical system corresponding to this potential. Draw the corresponding phase diagram. Justify.

[1] It is also useful to plot the potential using a computer, for various values of k and a

 End of Practice Problems

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B.Tech. Physics: Autumn 2023

Dia: Today! March 2023 **PH1L001** Time: três horas

Practice Problems - II :
Action Principle, Lagrangian Dynamics and Central Forces

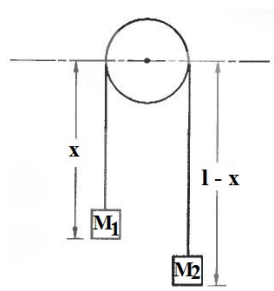
MM:80

1. Find the shortest distance between two points in the following cases below, using the action principle:
   (a) Two points in a plane (5)
   (b) Two points on the surface of earth (assuming earth to be a perfect sphere). (5)

2. Imagine a metal bead with a wire threaded through a hole in it, so that the bead can slide with no friction along the wire. How can one choose the shape of the wire so that the time of descent under gravity (from rest) is smallest possible[1]? Plot your solutions (*numerically*) and discuss the trajectory of the metal bead. (5)

3. Find the Euler-Lagrange equations for the following Lagrangians:
   (a) Simple Harmonic Oscillator: $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$ (5)
   (b) Projectile motion (with out air resistance): $L = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2\right) - mgy$ (5)
   (c) Particle in 2-dimensions (Polar coordinates): $L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - U(r)$ (5)
   (d) The Simple Pendulum: $L = \frac{1}{2}ml^2\dot{\theta}^2 + mgl\cos\theta$ (5)

4. Consider two masses M_1 and M_2 connected by a string (massless and inextensible) of length l passing over a pully (frictionless and massless)[2].



   (a) Find the Lagrangian and Euler-Lagrange equations of motion. Interpret the motion. (5)

[1] The problem of the determining the brachistochrone (shortest-time curve) was formally posed by Johann Bernouilli in 1696 as a challenge to the mathematicians of his day. Bernouilli himself found the solution (using a physical argument partly suggested by Fermat's "least-time" derivation of Snell's law of refraction in geometrical optics), and Newton gave a different proof. This problem is widely regarded as the founding problem of the 'calculus of variations' (finding the curve, or surface, minimising a given integral). This is the spirit of the approach developed by L.Euler (in 1736) and J-L. Lagrange (in 1755) to deal with general problems of this kind.

[2] This is known as an Atwood's machine

5. For two particles of masses m_1 and m_2 moving under the influence of a central potential $V(r)$, the motion always takes place in a plane. For the following two potentials: (i) $F = -k\,r$, $\quad (ii)$ $F = -k/r$.

   (a) Write the Lagrangian in a plane, identify the conserved quantities and obtain the Euler-Lagrange Equations of motion[3]. (5)

   (b) Identify the equivalent one-dimensional problem and the effective potential (5)

   (c) Plot the effective potential and discuss qualitatively all possible orbits of the particles. (5)

6. The orbit equation is given as[4]

$$\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right) + \frac{1}{r} = -\frac{\mu r^2}{l^2}\,F(r)$$

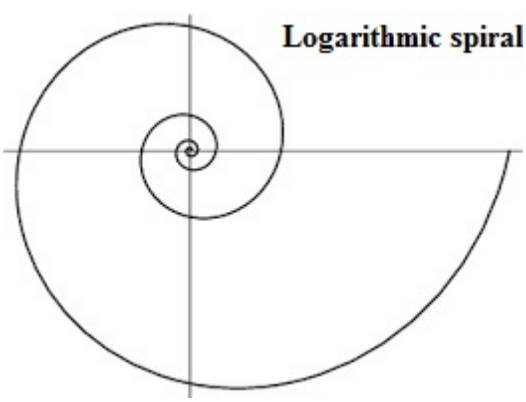
   (a) Find the force law for a particle to move in a logarithmic spiral orbit give by $r = k\,e^{\alpha\theta}$, where, k and α are constants. Determine $r(t)$ and $\theta(t)$. (5)

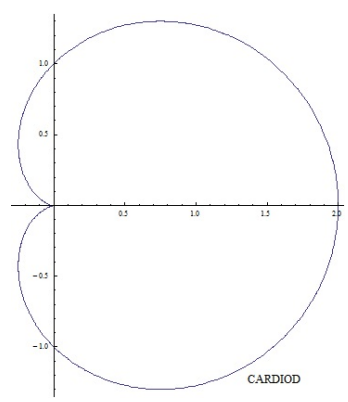
   (b) Find the force law for a particle in a circular orbit given by $r =$ constant. (5)

   (c) Find the force law for a central-force field that allows a particle to move in a spiral orbit given by $r = k\,\theta^2$, where k is a constant. (5)

   (d) Find the force law for particle in a cardiod orbit $r = k(1 + \cos\theta)$. (5)

   (e) Give numerical or polar plots of r vs. θ of all the above cases, $(a), (b), (c)$ and (d). Sample plots of spiral and cardiod are given below (generated through Mathematica software). (5)





[3]Solutions can be found numerically as well. It is a good exercise to numerically integrate the equations of motion

[4]The orbit equation can obtained after some algebra, from the Euler-Lagrange equations. Refer to the discussion in the class

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Date: Today! March 2023 **PH1L001** Time: uma hora

TUTORIAL - I : Review of Newtonian Mechanics MM:20

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1. (a) Find the displacement and velocity of a particle in horizontal motion in a medium in which the retarding force is proportional to the velocity. (5)

   (b) Find the displacement and velocity of a particle undergoing vertical motion in a medium in which the retarding force is proportional to the velocity. (5)

2. Consider, projectile motion in two dimensions. Let the muzzle velocity of the projectile be v_0 and the angle of elevation be θ. Calculate the projectile's displacement, velocity and range. (5)

   What happens when we add air resistance to the motion of the projectile[1]? Calculate the decrease in range, under the assumption that the force caused by the air resistance is directly proportional to the projectile's velocity[2].

3. Consider a projectile fired vertically in a constant gravitational field. For the same initial velocities, compare the times required for the projectile to reach its maximum height in the following two cases: (5)
   (a) for zero resisting force
   (b) for a resisting force proportional to the instantaneous velocity of the projectile.

4. **Bonus Question:** Consider a charged particle entering a region of uniform magnetic field **B** (for example, the earth's field). Set up the force balance equations, find the most general solution and discuss the subsequent motion of the particle[3].

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[1] *Curious case of Bumblebee:* There was a folklore surrounding the flight of Bumblebee (Not the autobots in Transformers!). It was believed by many people that Bumblebee is aerodynamically incapable of flying. Of course, Bumblebees did not know that and they continued to fly ! How did considerations of air resistance resolve the paradox?

[2] This problem may be attempted by numerical methods

[3] For simplicity, you can assume magnetic field to be only in y-direction.

End of Tutorial

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Dia: Oje! Março 2023 **PH1L001** Time: uma hora

Tutorial - II : Action Principle and Lagrangian Dynamics MM:20

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1. Consider a one-dimensional system with the following Lagrangian

$$L = e^{\gamma t}\left(\frac{m\dot{q}^2}{2} - V(q)\right)$$

where γ is a positive constant.

   (a) Find the Equations of Motion.

   (b) Using the above Lagrangian, explain what type of physical systems can be studied for the following choices of potentials $V(q) = \frac{k_1}{2}q^2$ and $V(q) = k_2 q$, where, k_1 and k_2 are arbitrary constants.

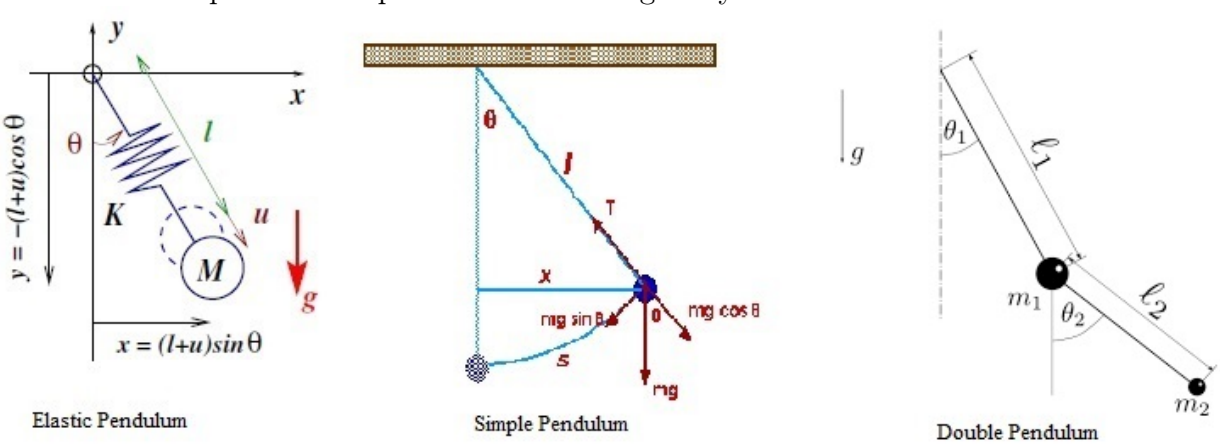
2. Identify the generalised coordinates, Lagrangian and compute the Euler-Lagrange equations.

   (a) A simple pendulum of mass m attached to a string of length l.

   (b) A simple pendulum of mass m attached to a string of length l, where, after the pendulum is set in to motion, the length of the string is shortened at a constant rate $\frac{dl}{dt} = \alpha$ (suspension point remains fixed). Is Energy conserved?

   (c) Elastic Pendulum: Mass m at the end of a spring, which can stretch. Length of the unstretched spring is l. u represents the stretch of the string. (5)

   (d) Double Pendulum: This is formed using two rigid, slender rods each of mass m and length l as shown in the figure below. Link 1 is free to pivot about point O and link 2 is free to pivot about point P. Note that gravity acts. (5)



Elastic Pendulum Simple Pendulum Double Pendulum

 End of Tutorial

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Dia: Oje! Março 2023 **PH1L001** Time: uma hora

Tutorial-cum-Practice Problems - III : Special Relativity MM:25

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1. Lorentz transformations in special relativity are given as

$$\begin{aligned} x' &= \gamma\,(x - vt) \\ y' &= y \\ z' &= z \\ t' &= \gamma\,(t - \delta) \end{aligned}$$

Here, (t, x, y, z) are coordinates of the observer O at rest, where as (t', x', y', z') are coordinates of the observer O' moving with uniform velocity v.

   (a) Derive and show that $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$ and $\delta = \frac{x\,v}{c^2}$ using Einstein's postulate that velocity of light is a constant. (5)

   (b) If the velocity components of a particle measured by a static observer O are[1] $u_x = \dot{x}, u_y = \dot{y}, u_z = \dot{z}$, then show that the velocity components of the particle measured by an observer O' (moving with velocity v) are $u'_x = \frac{u_x - v}{1-\frac{v}{c^2}u_x}$, $u'_y = \frac{u_y}{\gamma(1-\frac{v}{c^2}u_x)}$, $u'_z = \frac{u_z}{\gamma(1-\frac{v}{c^2}u_x)}$. (5)

2. The line element in special relativity to measure space-time distances is given as, $ds^2 = -c^2\,dt^2 + dx^2 + dy^2 + dz^2$.

   (a) Use the above line element to define a light-cone. Draw a light-cone diagram and mark the time-like, space-like and light-like regions clearly. (5)

   (b) Explain time-dilation by drawing a space-time diagram of observer O (at rest) and observer O' (moving with velocity v in positive x-direction). (5)

3. **Einstein's Mass-Energy equivalence:** Consider a particle of rest mass m_0, initially stationary at $x = x_i$. A force of magnitude F is then applied in the positive x-direction and the particle moves under the influence of the force. Show that the total work done by the force when the particle moves to position x_f is (5)

$$K(v) = \int_{x_i}^{x_f} F\,dx = E(v) - E(0)\,,$$

where $E(v) = mc^2$ and $E(0) = m_0c^2$, with $m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$.

[1] where $\dot{x} = \frac{dx}{dt}, \dot{y} = \frac{dy}{dt}, \dot{z} = \frac{dz}{dt}$.

 End of Tutorial